

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1340

Lone Tree 138 kV
June 15, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1340. The Interconnection Customer proposes a 203.9 MW solar photovoltaic facility interconnecting at the Lone Tree 138 kV substation in the Virginia area. The request was received on November 23, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 406 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,381,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Osceola-Michigan Power Lp 138kV Line	Craven County Wood Energy Lp-Lone Tree Substation 138kV Line	122.5	431	100.0
Lowmoor-Craven County Wood Energy Lp 138kV	Osceola-Lone Tree 138kV	122.2	433	100.0
Lone Tree-Osceola 138kV	Craven County Wood Energy Lp-Lone Tree 138kV L	122.1	437	100.0
Craven County Wood Energy Lp-Lone Tree 138kV line	Lone Tree-Osceola 138kV	121.9	438	100.0
Cloverdale Substation-Hondo 138kV line	Blue Ridge-Neely 138kV	101.2	195	15.7
Fudge Hollow Substation-Imhoff Tap 1 138kV	Aspen-Lowmoor 345kV	100.5	183	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Michigan Power Lp-Osceola 138kV	Rebuild line with higher-capacity conductor	\$10,751,000	37% (\$4,025,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lowmoor-Craven County Wood Energy Lp 138kV	Rebuild line with higher-capacity conductor	\$19,743,000	37% (\$7,391,000)
Lone Tree-Osceola 138kV L	Rebuild line with higher-capacity conductor	\$7,138,000	37% (\$2,672,000)
Lone Tree-Craven County Wood Energy Lp 138kV	Rebuild line with higher-capacity conductor	\$17,579,000	37% (\$6,581,000)
Cloverdale-Hondo Substation 138kV	Reconductor line	\$918,000	14% (\$125,000)
Fudge Hollow Substation-Imhoff Tap 1 138kV	Reconductor line	\$2,462,000	24% (\$587,000)
	Total allocated to project		\$21,381,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.